

SMART FINANCIAL SERVICES KIOSK WITH AI ADVISORY SYSTEM

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Project Abstract

The **Smart Financial Services Kiosk with AI Advisory System** integrates AI Financial Advisor, KYC, Fraud Detection, and User History into a single web-based platform. Developed using **Python, Flask, and Raspberry Pi 4 Model B**, the system provides secure financial guidance and digital document processing through an interactive interface. It improves accessibility, efficiency, and security while demonstrating AI-based financial service automation.

Problem Statement

Financial services are often spread across multiple platforms, making financial guidance, KYC verification, and fraud detection time-consuming and inconvenient. This project addresses these challenges by integrating all essential financial services into a single AI-powered web-based kiosk.

Need for the Project

The increasing demand for digital financial services requires a secure and user-friendly platform. This project integrates **AI Financial Advisory, KYC, and Fraud Detection** into a single web-based kiosk, making financial services more efficient and accessible.

Project Objectives

- Develop a **Smart Financial Services Kiosk** for integrated financial services.
- Implement **AI Financial Advisory** with secure **KYC verification**.
- Enhance service security through **Fraud Detection** and **User History Management**.

Expected Learning Outcomes

- Gain practical experience in **AI, Raspberry Pi, and Web Application Development**.
- Develop skills in **KYC Processing, Fraud Detection, and Financial System Integration**.

System Requirements

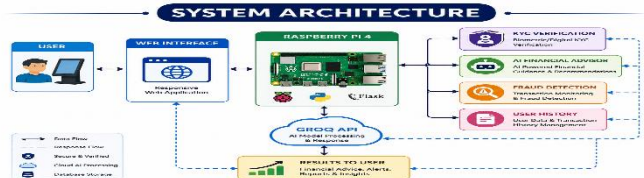
Hardware: The system is built using a **Raspberry Pi 4 Model B (4 GB RAM)**, **Samsung Galaxy Tab A11**, **32 GB microSD Card**, **Raspberry Pi Power Adapter**, **Network Connectivity**, **Connecting Cables**, and a **Kiosk Structure**.

Software: The application is developed using **Raspberry Pi OS**, **Python**, **Flask**, **HTML**, **CSS**, **Groq API**, **Groq Python Package**, **Python-dotenv**, **Requests**, and a **Python Virtual Environment**.

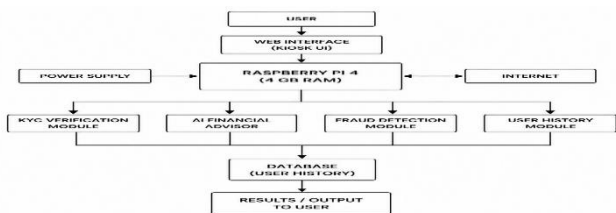
Methodology / Workflow

Requirement Analysis → System Design → Web Application Development → KYC Verification → AI Financial Advisory → Fraud Detection → Testing & Validation → Deployment

System Architecture



Block Diagram



Project Development Process

Problem Identification → Requirement Analysis → System Design → Hardware & Software Development → AI Module Integration → Testing & Validation → Final Deployment

Implementation

The system was implemented using **Raspberry Pi 4**, **Python**, and the **Flask** framework to develop a web-based financial kiosk. **Groq API** was integrated for AI-based financial advisory, while **KYC verification**, **Fraud Detection**, and **User History** modules were developed and tested to ensure secure and efficient financial services.

Algorithms / Models Used

- Algorithm / Model 1:** Groq Large Language Model (LLM) – Used for the **AI Financial Advisor** to generate financial guidance from user queries.
- Algorithm / Model 2:** Rule-Based Fraud Detection Model – Used to evaluate user-provided information and identify potential fraud risk indicators.



QR Code — Project
Report (PDF) | Source
Code Repository |
Demonstration Video

Dataset / Input Data (if applicable)

The system uses **financial queries, KYC documents, and fraud details** entered by users through the web interface for secure processing and AI-based analysis.

Experimental Setup

The project was tested using **Raspberry Pi 4 Model B**, **Samsung Galaxy Tab A11**, **Python**, **Flask**, **Groq API**, and **Raspberry Pi OS**. Experiments were conducted over a **Wi-Fi network** to validate the AI Financial Advisor, KYC processing, Fraud Detection, and User History modules.

Results

Module	Result
AI Financial Advisor	Successfully generated financial guidance
KYC Verification	Documents uploaded and processed successfully
Fraud Detection	Risk evaluation completed successfully
User History	User activities stored and retrieved successfully

Performance Evaluation

Metric	Target Performance
Login Authentication Accuracy	≥ 95%
KYC Verification Success Rate	≥ 98%
AI Financial Advisory Accuracy	≥ 90%
Fraud Detection Accuracy	≥ 95%
Dashboard Response Time	< 3 s
System Availability	≥ 99%

Testing and Validation

The system was tested for **AI Financial Advisory, KYC Verification, Fraud Detection, User History, and overall performance** to ensure secure and reliable operation.

Challenges Encountered

- Challenge 1:** Biometric hardware integration.
- Challenge 2:** Low-quality KYC documents.

Solutions Implemented

- Solution 1:** Multi-factor authentication implemented.
- Solution 2:** Document validation for KYC verification.

Innovation / Novel Features

The system integrates **AI Financial Advisory, Digital KYC, Fraud Detection, and User History** into a single **secure web-based platform** for efficient financial services.

Applications

- Banks and Financial Institutions**
- Government Service Centers (Digital Financial Kiosks)**

Future Scope

Future enhancements include **advanced AI, OCR-based KYC, blockchain security, and mobile/cloud integration** for smarter and more secure financial services.

Project Outcomes

The project developed a **web-based Smart Financial Services Kiosk** with **AI Financial Advisory, Digital KYC, Fraud Detection, and User History**, providing secure and efficient digital financial services.

Skills Acquired

- Technical Skill:** Python, Flask, Raspberry Pi, AI Integration
- Soft Skill:** Teamwork, Problem Solving, Communication, Project Management

References

- Intelligent Finance: The Evolution and Impact of AI-Driven Advisory Services in Fintech, Journal, 2021.
- Kiosk Banking: The Challenges to and Implications of Building an Inclusive Financial Infrastructure, Journal, 2019
- Knowledge Institute of Technology, Department of Mechanical Engineering, Salem, Tamil Nadu, India.

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